



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 55



KUKJE RICE TRANSPLANTER Model : RR-4B

KUKJE RR-4B RICE TRANSPLANTER

Test Report No.	:	2013 - 55
Date of Test	:	June 22, 2013
Test Valid Until	:	June 22, 2018
Brand	:	KUKJE
Model	:	RR-4B
Type	:	Walk-behind, 4-row, power-operated mechanical rice transplanter
Manufacturer	:	KUKJE MACHINERY CO., LTD. 11-1 Yangsu-ri Okcheon-eup, Okcheon-gun, Chungbuk, South Korea
Test Requested by	:	ACT MACHINERIES AND METALCRAFT CORPORATION Turayong, Cauayan City, Isabela, and FIT COREA TRADING PHILS., INC. #38-C West Triangle, West Avenue, Quezon City
Machine Classification	:	Commercial

SUMMARY

The KUKJE RR-4B Rice Transplanter was tested for actual field performance. It was tested using seedlings and field area specially prepared for the transplanter using the company's recommended procedure for seedlings and field area preparation.

The average actual field capacity of the machine was 0.157 ha/h with field efficiency of 73.3% and 3.83% missed hills. The average forward speed was 2.12 kilometers per hour (kph). The fuel consumption of the 2.60 kW gasoline engine was 0.81 L/h.

The average distance between hills was 175 mm with variation of ± 1.49 and error of 7.08% based on machine's distance between hills setting of 180 mm. The average distance between rows was 300 mm. The average depth of planting was 56 mm with variation of ± 4.64 . The average number of seedlings per hill was six with variation of ± 1.36 and percent error of 23.9% based on machines number of seedlings per hill setting of eight.

The floating seedlings and buried seedlings were both 1.43% and no damaged seedlings.

No breakdown or any malfunctioning occurred during the test.

PURPOSE AND SCOPE OF TEST

The rice transplanter was tested for verification of performance. The test was conducted to verify and assess the following:

1. Actual Field Capacity
2. Field Efficiency
3. Operating accuracy and suitability
 - a. Distance between hills and rows
 - b. Number of seedlings per hill
 - c. Depth of seedling
 - d. Percentage missed hills, floating seedlings, buried seedlings, and damaged seedlings
4. Fuel Consumption

PLACE AND DATE OF TEST

The KUKJE RR-4B Rice Transplanter was tested at the Badua's Rice Farm in Barangay Bagong Sikat, Muñoz, Nueva Ecija on June 22, 2013.

OPERATOR'S MANUAL

The operator's manual submitted with the machines contains the following information: Safety Precautions, Raising condition of young rice and field condition, Controls and components, Description of operating system, Working with the machine, Machine care after working, Periodic check and maintenance, Troubleshooting, and Specification.

DESCRIPTION OF THE IMPLEMENT

The KUKJE RR-4B Rice Transplanter was a walk-behind type, 4-row, power-operated mechanical rice transplanter (**Fig. 1**). It consisted of the following main components: the prime mover, planting mechanism, wheels and floats, and operating controls.

The prime mover (**Fig. 2**) was a 2.60 kW four-stroke, air-cooled OHV gasoline engine integrally mounted at the front most side of the machine. It was the source of all the power of the transplanter.

The planting mechanism was a four-row planting fork (**Fig. 3**) which picks the seedlings through a reciprocating four compartment trays accessible to the operator. It also had spare seedling stand (**Fig. 4**) located on the top most portion of the machine which was also accessible to the operator.

It had two wheels with paddles (**Fig. 5**) used for moving the machine and floats (**Fig. 6a**) used to prevent the machine from sinking in the muddy field.

The Main operation controls (**Fig. 6b**) were located at the handle bars which were accessible to the operator. Also beneath the handle bars was the planting quantity controls (**Fig.7**).

Detailed specifications of the machine are shown in **Table 1**.



Figure 1. KUKJE Rice Transplanter
Model: RR-4B



Figure 2. Prime mover, a 2.60 kW gasoline engine

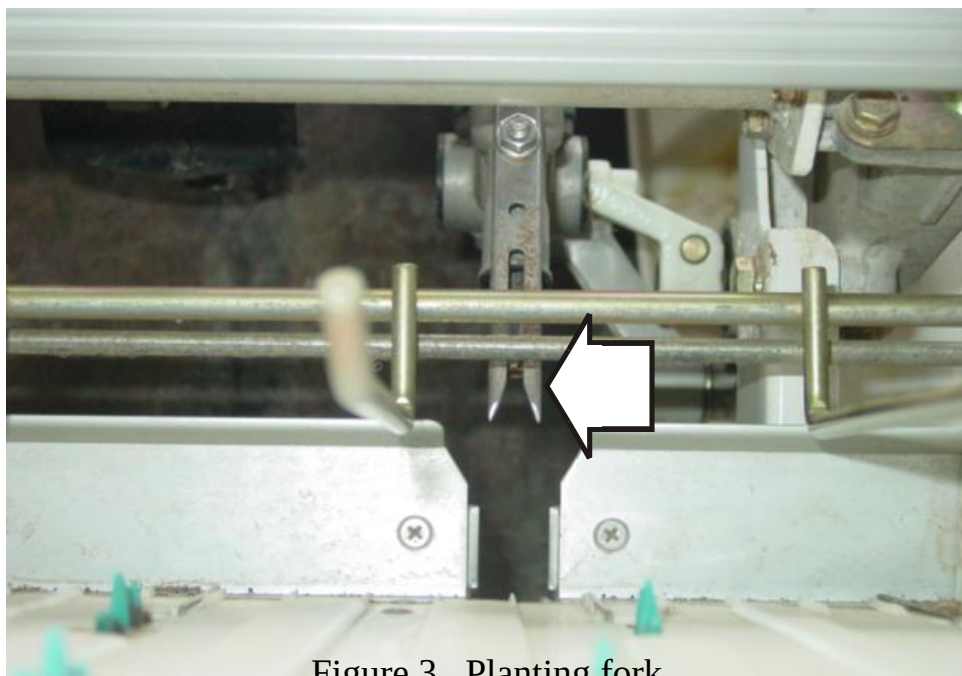


Figure 3. Planting fork

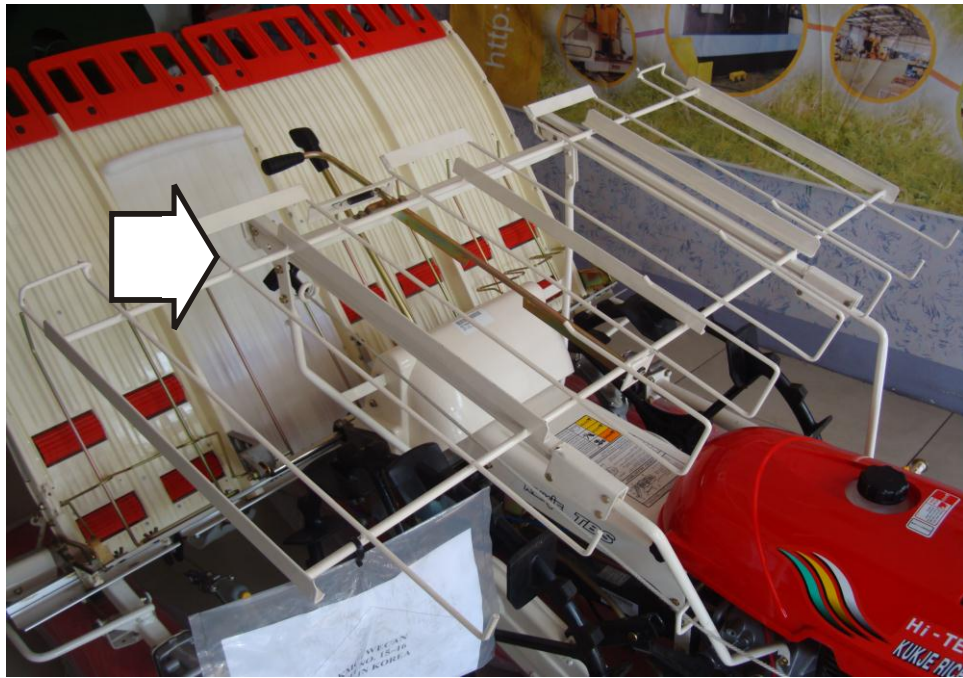


Figure 4. Spare seedling stand



Figure 5. Wheel with paddles

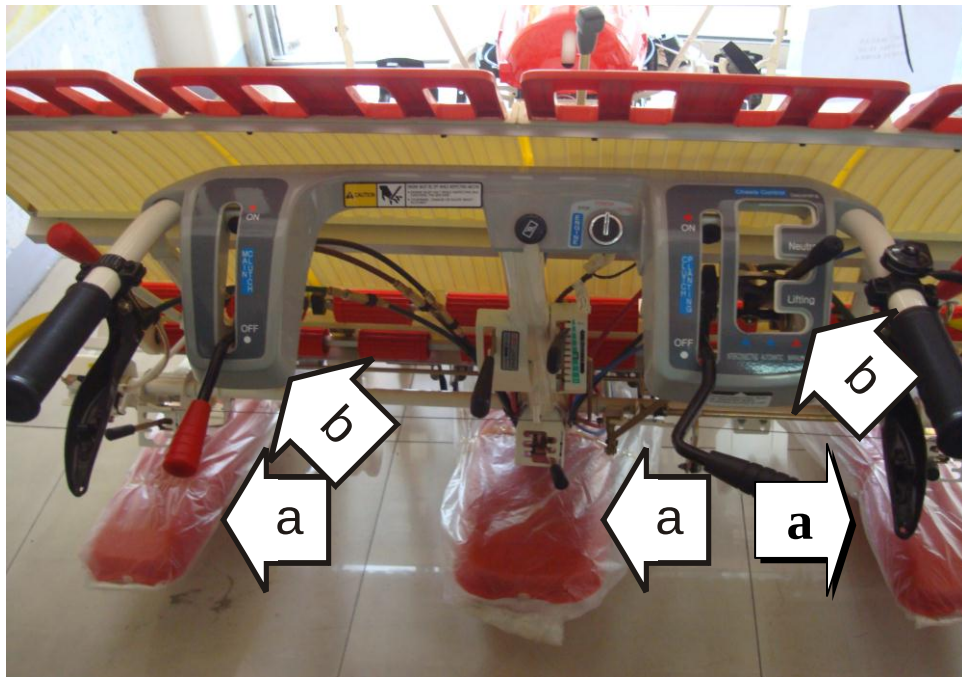


Figure 6. a) Transplanter Floats
b) Main operation controls



Figure 7. Planting Quantity Control

MACHINE SPECIFICATIONS

Table 1. Detailed specifications of the KUKJE RR-4B Rice Transplanter

1.	Brand	:	KUKJE
2.	Model	:	RR-4B
3.	Serial No.	:	-
4	Manufacturer	:	KUKJE MACHINERY CO., LTD. 11-1 Yangsu-ri Okcheon-eup, Okcheon-gun, Chung-buk, South Korea
5.	Dimensions and weight		
5.1	Overall length (mm)	:	2200
5.2	Overall width (mm)	:	1500
5.3	Overall height (mm)	:	1000
5.4	Weight (kg)	:	No data available
6.	No. of planting row	:	4
7.	Row Distance (mm)	:	300
8.	Spare seedling stand		
8.1	Size, L x W (mm)	:	1230 x 855
8.2	Location	:	Above paddle wheels
9.	Paddle wheels		
9.1	Quantity	:	2
9.2	Size, D x W (mm)	:	600 x 85
9.3	No. of paddles/wheel	:	9

Table 1. Detailed specifications of the KUKJE RR-4B Rice Transplanter (cont'n)

10.	Engine		
10.1	Brand	:	ROBIN-SUBARU
10.2	Model	:	EY15-2B
10.3	Type	:	Four-stroke, air-cooled, OHV gasoline engine
10.4	Rated Output Power (kW)		
	Maximum	:	2.60 @ 2000 rpm
	Continuous	:	2.0 @ 2000 rpm
10.5	Starting System	:	Rope (Recoil System)

RESULTS AND DISCUSSIONS

Results of performance tests are shown in **Table 2-5**.

Test Seedlings

The seedlings used were prepared based on the recommended seedlings preparation of the company specific to the transplanter. It uses a soil-bearing type of seedlings raised in seedling trays designed for the transplanter.

Performance

The performance of the rice transplanter was evaluated in terms of its field capacity, field efficiency, missed hills, floating seedlings, buried seedlings and damaged seedlings.

Actual Field Capacity. It is the ratio of the area planted and the total time of operation. The Field Capacity of the machine was 0.157 ha/h or 6.37 h/ha.

Field Efficiency. It is the ratio of the Actual Field Capacity and the Theoretical Field Capacity, expressed in percent. The Field Efficiency of the machine was 73.3%.

Missed Hills. This refers to the number of strokes the transplanter had made but was not able to pick the seedlings. The average percentage missed hills was 3.83%.

Floating Seedlings. This refers to the seedlings that were picked out by the transplanter picker during operation but were not able to settle it in the soil for reasons such as high water level, unlevelled soil, population of grown seedlings, etc. The average floating seedlings were 1.43%.

Buried Seedlings. This refers to the seedlings that were stuck under the mud. The average buried seedlings were 1.43%.

Damaged Seedlings. This refers to the seedlings which were crushed, cut or bent due to the machine. It can be caused by cutting or bending of the seedlings during the process of picking or planting by the planting fork or pincette. There were no damaged seedlings.

Performance Based on Machine's Setting

The average distance between hills was 175 mm with variation of $\pm 1.49\%$ and error of 7.08% based on machine's distance between hills setting of 160 mm. The average distance between rows was 300 mm. The average depth of planting was 56 mm with variation of ± 4.64 . The average number of seedlings per hill was six with variation of $\pm 1.36\%$ and error of 23.9% based on machines number of seedlings per hill setting of eight.

Fuel Consumption

The 2.60 kW ROBIN-SUBARU EY15-2B Gasoline engine prime mover had a Fuel Consumption Rate of 0.81 L/h.

Labor Requirement

One person was needed to operate the rice transplanter (**Figure 8**).

OBSERVATIONS

1. The rice transplanter was easy to turn at corners (**Figure 9**).
2. The machine was provided with controls and guides for ease of operation.
However, the quality of work (**Figure 10**) was based on the skill of the operator.
3. The rice transplanter was provided with informative stickers (**Figure 11**).



Figure 8. Operator of the rice transplanter



Figure 9. Turning at corner



Figure 10. Quality of work



Figure 11. Sticker attached to the rice transplanter

FIELD PERFORMANCE TEST

Machine Tested : KUKJE Rice Transplanter
Model: RR-4B

Location of Test : Badua's Farm, Brgy. Bagong Sikat,
Muñoz, Nueva Ecija

Date of Test : June 22, 2013

Table 2. Seedling Conditions

ITEMS	Data
1. Variety	Diamond X
2. Days After Seeding	14
3. Seedling Height (mm)	191

Table 3. Field Conditions

ITEMS	Data
1. Field Dimension, L x W (m)	-
2. Area (m ²)	1326.6
3. Soil Type	Clay Loam
4. Soil Resistance (kg/cm ²)	No data

Table 4. Machine Settings

ITEMS	Data
1. Distance between rows (mm)	300
2. Distance between hills (mm)	180
3. No. of seedlings per hill	8
4. Position of Float Control (Driving, Planting)	Planting

Table 5. Results of Performance Test

ITEMS	Data
1. Total Time of Operation (h)	0.875
2. Effective Working Width (mm)	1156
3. Travelling Speed (kph)	2.12
4. Field Capacity a) ha/h b) h/ha	0.157 6.37
5. Theoretical Field Capacity a) ha/h b) h/ha	0.215 4.65
6. Field Efficiency (%)	73.3
7. Distance Between Rows (mm)	300
8. Distance Between Hills (mm) a) Variation (%) b) Percent Error (at setting of 180 mm)	175 ± 1.49 7.08
9. Number of Seedlings per Hill a) Variation (%) b) Percent Error (at setting of 8 seedlings/hill)	6 ± 1.36 23.9
10. Depth of Planting (mm) a) Variation (%)	56 ± 4.64
11. Missed Hills (%)	3.83
12. Damaged Seedlings (%)	Nil
13. Buried Seedlings (%)	1.43
14. Floating Seedlings (%)	1.43
15. Fuel Consumption (L/h)	0.81

Test Engineer:

ROMULO E. EUSEBIO

Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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